

Lines and Planes in Three Dimensions

ANSWER KEY



The vector equation of a line

- 1 Write the vector equation of the line through the point $(2, -1, 3)$ with direction vector $(1, 2, -1)$.
 $\mathbf{r} = (2, -1, 3) + t(1, 2, -1)$, where $t \in \mathbb{R}$.
- 2 Find the vector equation of the line through the two points $A(1, 0, 2)$ and $B(4, 3, 1)$.
Direction vector: $\vec{AB} = (4 - 1, 3 - 0, 1 - 2) = (3, 3, -1)$. Vector equation: $\mathbf{r} = (1, 0, 2) + t(3, 3, -1)$.

Determining if two lines intersect

- 3 Determine whether the lines $\mathbf{r}_1 = (1, 2, 0) + s(1, -1, 2)$ and $\mathbf{r}_2 = (0, 3, 1) + t(2, 1, -1)$ intersect, and if so, find the point of intersection.
Equate components: $1 + s = 2t$, $2 - s = 3 + t$, $2s = 1 - t$. From the second: $s = -1 - t$. Substitute into the first: $1 + (-1 - t) = 2t$, so $-t = 2t$, giving $t = 0$, then $s = -1$. Check the third equation: $2(-1) = 1 - 0$? $-2 = 1$ — false. The lines do not intersect (they are skew, since they are not parallel either).

The equation of a plane

- 4 Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.
The plane has the form $2x - y + 3z = d$. Substituting the point: $2(1) - 2 + 3(-1) = 2 - 2 - 3 = -3$.
Equation: $2x - y + 3z = -3$.
- 5 Find the Cartesian equation of the plane containing the points $A(1, 0, 0)$, $B(0, 1, 0)$, and $C(0, 0, 1)$.
 $\vec{AB} = (-1, 1, 0)$, $\vec{AC} = (-1, 0, 1)$. Normal
 $= \vec{AB} \times \vec{AC} = (1 \cdot 1 - 0 \cdot 0, 0 \cdot (-1) - (-1) \cdot 1, (-1) \cdot 0 - 1 \cdot (-1)) = (1, 1, 1)$. Plane: $x + y + z = d$;
substituting A: $1 + 0 + 0 = 1$. Equation: $x + y + z = 1$.

Angle between a line and a plane

- 6 Find the angle between the line with direction $(1, 2, 2)$ and the plane $x + y + z = 5$, to the nearest degree.
Plane normal: $\mathbf{n} = (1, 1, 1)$. $\mathbf{d} \cdot \mathbf{n} = 1 + 2 + 2 = 5$. $|\mathbf{d}| = 3$, $|\mathbf{n}| = \sqrt{3}$. $\sin \theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|} = \frac{5}{3\sqrt{3}} \approx 0.962$,
so $\theta \approx 74^\circ$.

Distance from a point to a plane

- 7 Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using

$$d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}$$
$$d = \frac{|2(3) - 1(1) + 2(-2) - 6|}{\sqrt{4 + 1 + 4}} = \frac{|6 - 1 - 4 - 6|}{3} = \frac{|-5|}{3} = \frac{5}{3}.$$

Finding the angle between two planes

- 8 Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.

Normals: $\mathbf{n}_1 = (1, 2, -1)$, $\mathbf{n}_2 = (2, -1, 1)$. $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 2 - 1 = -1$. $|\mathbf{n}_1| = \sqrt{6}$, $|\mathbf{n}_2| = \sqrt{6}$.

$$\cos \theta = \frac{|-1|}{\sqrt{6} \cdot \sqrt{6}} = \frac{1}{6}, \text{ so } \theta \approx 80^\circ.$$

Finding the intersection line of two planes

- 9 Find a vector equation for the line of intersection of the planes $x + y + z = 6$ and $x - y + 2z = 5$, using the cross product of the two normal vectors as the direction, and finding one point on both planes by inspection.

Direction:

$$\mathbf{n}_1 \times \mathbf{n}_2 = (1, 1, 1) \times (1, -1, 2) = (1 \cdot 2 - 1 \cdot (-1), 1 \cdot 1 - 1 \cdot 2, 1 \cdot (-1) - 1 \cdot 1) = (3, -1, -2).$$

To find a point, try $z = 0$: $x + y = 6$ and $x - y = 5$, giving $x = 5.5$, $y = 0.5$. Point: $(5.5, 0.5, 0)$. Line:

$$\mathbf{r} = (5.5, 0.5, 0) + t(3, -1, -2).$$

Testing whether a line lies within a plane

- 10 Determine whether the line $\mathbf{r} = (1, 2, 3) + t(2, -1, 1)$ lies entirely within the plane $x + 3y + z = 10$.

A line lies within a plane if (1) its direction vector is perpendicular to the plane's normal, and (2) a point on the line satisfies the plane's equation. Normal of the plane: $(1, 3, 1)$. Direction of line: $(2, -1, 1)$. Dot product: $1(2) + 3(-1) + 1(1) = 2 - 3 + 1 = 0$ ✓ (perpendicular to normal, so parallel to the plane).

Check the point $(1, 2, 3)$: $1 + 3(2) + 3 = 1 + 6 + 3 = 10$ ✓ (satisfies the plane equation). Since both conditions hold, the line lies entirely within the plane.

Finding the point where a line meets a plane

- 11 Find the point where the line $\mathbf{r} = (1, 0, 2) + t(1, 2, -1)$ intersects the plane $2x + y - z = 3$.

Substitute the parametric coordinates $x = 1 + t$, $y = 2t$, $z = 2 - t$ into the plane equation:

$$2(1 + t) + 2t - (2 - t) = 3. \text{ Expand: } 2 + 2t + 2t - 2 + t = 3, \text{ so } 5t = 3, \text{ giving } t = \frac{3}{5}.$$

Substituting back: $x = 1 + \frac{3}{5} = \frac{8}{5}$, $y = 2\left(\frac{3}{5}\right) = \frac{6}{5}$, $z = 2 - \frac{3}{5} = \frac{7}{5}$. Point of intersection: $\left(\frac{8}{5}, \frac{6}{5}, \frac{7}{5}\right)$.

Shortest distance between two skew lines

- 12 Two skew lines have direction vectors $\mathbf{d}_1 = (1, 0, 1)$ and $\mathbf{d}_2 = (0, 1, 1)$, and pass through the points $A(0, 0, 0)$ and $B(1, 1, 0)$ respectively. Find the shortest distance between the two lines, using

$$d = \frac{|\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}.$$

$$\mathbf{d}_1 \times \mathbf{d}_2 = (0 \cdot 1 - 1 \cdot 1, 1 \cdot 0 - 1 \cdot 1, 1 \cdot 1 - 0 \cdot 0) = (-1, -1, 1). \vec{AB} = (1, 1, 0).$$

$$\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2) = 1(-1) + 1(-1) + 0(1) = -2. |\mathbf{d}_1 \times \mathbf{d}_2| = \sqrt{1 + 1 + 1} = \sqrt{3}. \text{ Distance} \\ = \frac{|-2|}{\sqrt{3}} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}.$$

Foot of the perpendicular from a point to a line

- 13 Find the foot of the perpendicular from the point $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$.

A general point on the line is $(1 + t, t, t)$. The vector from this point to P is

$(4 - (1 + t), 0 - t, 1 - t) = (3 - t, -t, 1 - t)$. For the foot of the perpendicular, this vector must be perpendicular to the direction $(1, 1, 1)$: $(3 - t)(1) + (-t)(1) + (1 - t)(1) = 0$, i.e. $3 - t - t + 1 - t = 0$, so $4 - 3t = 0$, giving $t = \frac{4}{3}$. Foot of perpendicular: $(1 + \frac{4}{3}, \frac{4}{3}, \frac{4}{3}) = (\frac{7}{3}, \frac{4}{3}, \frac{4}{3})$.

- 14 Determine whether the planes $2x - y + 3z = 4$ and $-4x + 2y - 6z = -8$ represent the same plane, a pair of parallel (distinct) planes, or intersecting planes.

The second equation's coefficients $(-4, 2, -6)$ are exactly -2 times the first equation's coefficients $(2, -1, 3)$, and the constant -8 is also -2 times 4 . Since the entire equation is a scalar multiple of the first, the two equations describe the same plane.

Checking whether a point lies on a plane

- 15 Determine whether the point $P(2, -1, 4)$ lies on the plane $3x + 2y - z = 1$, and if not, state whether it lies on the same side as the origin.
Substituting P : $3(2) + 2(-1) - 4 = 6 - 2 - 4 = 0 \neq 1$, so P does not lie on the plane. Substituting the origin $(0, 0, 0)$: $3(0) + 2(0) - 0 = 0$. Since the plane value at P (0) equals the plane value at the origin (0), and both are less than the required value of 1 , P lies on the same side of the plane as the origin.
- 16 Find the value of k such that the line $\mathbf{r} = (1, 2, 0) + t(2, -1, k)$ is parallel to the plane $x + 2y - z = 5$.
A line is parallel to a plane when its direction vector is perpendicular to the plane's normal vector. Normal of the plane: $(1, 2, -1)$. Direction of line: $(2, -1, k)$. Setting the dot product to zero:
 $1(2) + 2(-1) + (-1)(k) = 0$, so $2 - 2 - k = 0$, giving $k = 0$.
- 17 A surveyor models a hillside as the plane $2x + 3y + 6z = 42$, where x and y are horizontal coordinates (in metres) and z is height (in metres). Find the height z of the hillside at the point where $x = 6$ and $y = 4$, and find the plane's intercept on the z -axis (i.e. the height at $x = 0, y = 0$).
Substituting $x = 6, y = 4$: $2(6) + 3(4) + 6z = 42$, so $12 + 12 + 6z = 42$, giving $6z = 18$, so $z = 3$ metres. The z -intercept (at $x = 0, y = 0$): $6z = 42$, so $z = 7$ metres.

- 18 This question has three parts. Two lines are given by $\mathbf{r}_1 = (2, 0, 1) + s(1, 1, 0)$ and $\mathbf{r}_2 = (0, 2, 1) + t(1, -1, 0)$. (a) Show that both lines lie in the plane $z = 1$. (b) Find the point of intersection of the two lines. (c) Find the acute angle between the two lines, to the nearest degree.
- (a) Both lines have z -component fixed at 1 (since the direction vectors $(1, 1, 0)$ and $(1, -1, 0)$ have zero z -component, and both starting points have $z = 1$), so every point on either line satisfies $z = 1$; both lines lie in the plane $z = 1$. (b) Equating: $2 + s = t$ and $s = 2 - t$. Substituting the second into the first: $2 + (2 - t) = t$, so $4 - t = t$, giving $t = 2$, then $s = 0$. Point: $\mathbf{r}_1$ at $s = 0$ gives $(2, 0, 1)$; checking $\mathbf{r}_2$ at $t = 2$: $(0, 2, 1) + 2(1, -1, 0) = (2, 0, 1)$ ✓. Intersection point: $(2, 0, 1)$. (c) Direction vectors: $(1, 1, 0)$ and $(1, -1, 0)$. Dot product: $1 - 1 + 0 = 0$. Since the dot product is zero, the lines are perpendicular, so the angle between them is 90° .